WEEK 4 · LECTURE L3 · UNIT II

Designing the HR Technology Stack

Systems of record and engagement, the order of building, what success means, and privacy built in by design.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Distinguish systems of record, engagement and insight, credited as the course’s framing building on Moore (2011).
2. Map HR modules onto the employee lifecycle and sequence a stack by dependency and capacity.
3. Rationalize a system portfolio, and judge success with DeLone and McLean rather than by use alone.
4. Distinguish TOE from TOP, and apply India Stack, Privacy by Design and the DPDP Act to stack design.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 2 and CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

A Bengaluru IT services firm with 12,000 staff is building its HR stack from scratch. Which should it put in place first?

A

A leadership dashboard, so the Board sees progress early

B

A mobile app employees will enjoy using

C

The one system of record that every other module reads from

D

Whichever module has the cheapest license

EXAM LENS

The answer is C. Every engagement and insight module reads data that only the record can keep right.

Two roles, built for different things

	System of record	System of engagement
Purpose	Hold official data and transactions	Support how people interact and get things done
Built for	Accuracy, control, compliance	Ease, speed, reach
HR examples	Core HR, payroll, statutory records	Self service apps, chatbots, mobile onboarding
Typical users	HR and payroll specialists	Every employee and manager
Failure looks like	Wrong pay, wrong headcount, audit findings	Nobody uses it; work drifts back to email and spreadsheets

CITE CORRECTLY

The course adds a third role, insight, for analytics. Cite the three roles as the course's framing building on Moore (2011); they are not in Kavanagh and Johnson.

THE MODULE LANDSCAPE

Modules across the employee lifecycle

Stage	Modules	Role
Hire	ATS, referral portal	Engagement
Join	Onboarding, documents, eSign	Engagement
Work	Core HR, payroll, attendance	Record
Ask	Self service app, HR chatbot	Engagement
Grow	LMS, performance	Engagement
Listen	Pulse surveys, analytics	Insight

EXAM LENS

Every engagement and insight module reads employee data that only the record can keep right.

Course framing: modules from the Lesson Plan, Week 4, placed on the lifecycle and the three roles.

SEQUENCE BY DEPENDENCY

Record, then engagement, then insight

1 · Record

Every other module reads from it, so it comes first.

2 · Engagement

Mobile and vernacular channels decide who the stack actually reaches.

3 · Insight

Analytics on a split record produces confident errors, so it comes last.

EXAM LENS

Capacity binds as hard as money: each go live spends managers’ time and employees’ patience, and two large launches in one season overload frontline managers. Within each layer, sequence by the group that is worst served. A common reversal: analytics bought first to impress the Board, while two systems still disagree on headcount.

RATIONALIZING A PORTFOLIO

Four bins

Bin	Rule of thumb
Keep and integrate	Adoption 0.70 or more and 3 or more integrations
Migrate	Needed, but on premises or not mobile
Evaluate	Adoption 0.40 to 0.70, or a high cost per user
Retire	Adoption below 0.40 with a license cost, or no integrations

EXAM LENS

Four measures: adoption (active ÷ licensed), integrations, hosting and mobile reach, and cost per active user. The trap: a rule can retire what you need, such as a young pilot with low use, or the one channel the frontline already trusts.

Course framing: a rule of thumb to argue with, not a formula to obey.

USE IS NOT SUCCESS

High use, low satisfaction, no alternative

High use

Everyone opens it every day.

Low satisfaction

Everyone complains about it.

No alternative

It is the only thing that works.

EXAM LENS

A system everyone uses may simply be the one nobody can escape. Success has several dimensions; use is one of them, and it can be forced by the lack of any alternative (DeLone and McLean, 1992).

DELONE AND MCLEAN (2003)

What success means

System quality
Reliable, easy, fast.

Information quality
Accurate, complete, timely data.

Service quality
Support from HR and IT.

Use and intention to use
Voluntary or mandatory?

User satisfaction
The better signal when use is forced.

Net benefits
For employees, HR and the firm; they feed back into use and satisfaction.

EXAM LENS

Judge a stack on quality, use, satisfaction and benefits together, never on one of them alone.

TWO ADOPTION FRAMEWORKS WITH SIMILAR NAMES

TOE and TOP

	TOE (Tornatzky and Fleischer, 1990)	TOP (Bondarouk et al., 2017)
Contexts	Technology, organization, environment	Technology, organization, people
Question it answers	Why does a firm decide to adopt?	Why does an eHRM adoption succeed or fail inside the firm?
What it adds	Regulation, competitors, vendor market	Top management support, user acceptance, HR skills, culture
Use it for	The decision to invest in a stack	The rollout and its consequences

EXAM LENS

TOE explains the decision to adopt; TOP explains whether the adoption works.

BUILD LESS

India Stack as public rails

Identity

Aadhaar eKYC to verify who a new joiner is.

Documents

DigiLocker for verified certificates and records.

Signatures

eSign for offer letters and declarations.

Payments

UPI for stipends, reimbursements and payouts.

EXAM LENS

The question each rail raises: what is collected, for what stated purpose, kept for how long, and with whose consent? Public rails reduce what a firm must build, but not what it must answer for.

CAVOUKIAN (2011)

Privacy by Design: seven foundational principles

1
Proactive, not reactive; preventive, not remedial

2
Privacy as the default setting

3
Privacy embedded into design

4
Full functionality: positive sum, not zero sum

5
End to end security across the data lifecycle

6
Visibility and transparency

7
Respect for user privacy: keep it user centric

EXAM LENS

Built in at design, not bolted on after the data has moved.

What the stack must answer for

Lawful grounds

Consent, or a legitimate use (Sections 4, 6, 7).

Employment use

Section 7(i): processing for the purposes of employment.

Fiduciary duties

Notice, security safeguards, accuracy (Sections 5, 8).

Employee rights

Access, correction and erasure, grievance redress (Sections 11 to 13).

Data abroad

Transfers can be restricted to notified countries (Section 16).

Open question

“Purposes of employment” is undefined: does it reach applicants and former staff?

EXAM LENS

Employment purposes make many HR uses lawful without consent, but not every use, and not for every person. Other duties still apply.

APPLY

Designing a stack for a growing logistics firm

Teaching case (Course framing)

A Jaipur logistics firm with 4,000 staff, most of them drivers and warehouse workers with only a smartphone, runs payroll on a vendor system, keeps employee records in spreadsheets, and has just bought an analytics tool hosted abroad. The CEO wants a WhatsApp HR helpdesk next month.

Work it through

1. Map each system to record, engagement or insight. Which role is missing?
2. In what order should the firm build, and what should it pause?
3. How would you judge success in a year, using DeLone and McLean?
4. Which privacy questions must the analytics tool and the helpdesk answer first?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
The option that buys the most layers is the best option	Buy layers in dependency order: record, then engagement, then insight
A system everyone uses is a system that works	Use is one dimension of success; it can reflect the lack of any alternative
Employee data always needs consent under the DPDP Act	Section 7(i) makes employment purposes a legitimate use, but its reach is debated and other duties still apply
TOE and TOP are the same framework	TOE explains the decision; TOP explains the rollout

COMPARE AND EVALUATE

Six lenses on stack design

Lens	Use it to	Watch out for
Record, engagement, insight	Sequence the stack	The course’s framing, not Kavanagh’s
Portfolio bins	Rationalize systems	A rule can retire what you need *
DeLone and McLean (2003)	Define success	Mandatory use
TOE (1990)	Explain the decision to adopt	Factors without weights
TOP (2017)	Explain whether adoption works	Thin on strategic outcomes
Privacy by Design (2011)	Build privacy in	Principles, not procedures *

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

A Hyderabad hospital chain runs two HR systems, a WhatsApp helpdesk and an analytics tool bought last year. Using the three layer architecture framing, the DeLone and McLean model and Privacy by Design, propose a sequence for its HR technology stack and defend it against one objection. **20 marks**

A strong answer: maps systems to record, engagement and insight, credits the framing correctly, and names the layer that must come first; defines success beyond use; applies Privacy by Design and the DPDP Act; answers one objection, such as cost, lock in or parallel running.

“A system everyone uses is a successful system.” Evaluate this claim using DeLone and McLean and the idea of mandatory use. **20 marks**

A strong answer: explains the six dimensions; separates voluntary from mandatory use; uses satisfaction and net benefits as better signals; gives an HR example.

Distinguish the TOE and TOP frameworks and say when each is the right tool. **10 marks**

A strong answer: names each context set; TOE for the decision to invest, TOP for the rollout and its consequences.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Record, then engagement, then insight.**
   Dependency order, within money and capacity.

2. **Use is not success.**
   Judge quality, satisfaction and net benefits too.

3. **Public rails reduce building, not accountability.**
   India Stack still raises purpose, retention and consent questions.

4. **Privacy is built in at design.**
   Not bolted on after the data has moved.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Bondarouk, T., Parry, E., & Furtmueller, E. (2017).
- Johnson, R. D., Carlson, K. D., & Kavanagh, M. J. (2025), Chapters 1 and 2.

NEXT WEEK

- Scully-Russ and Torraco (2020), pp. 83 to 84; the Infosys Lex case; NASSCOM FutureSkills, selected sections.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *The International Journal of Human Resource Management, 28*(1), 98 to 131. https://doi.org/10.1080/09585192.2016.1245672
- Cavoukian, A. (2011). *Privacy by design: The 7 foundational principles* (revised). Information and Privacy Commissioner of Ontario.
- DeLone, W. H., & McLean, E. R. (1992). Information systems success: The quest for the dependent variable. *Information Systems Research, 3*(1), 60 to 95. https://doi.org/10.1287/isre.3.1.60
- DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten year update. *Journal of Management Information Systems, 19*(4), 9 to 30. https://doi.org/10.1080/07421222.2003.11045748
- Government of India. (2023). *The Digital Personal Data Protection Act, 2023* (Act No. 22 of 2023).
- Johnson, R. D., Carlson, K. D., & Kavanagh, M. J. (Eds.). (2025). *Human resource information systems* (6th ed.). Sage.
- Kavanagh, M. J., & Johnson, R. D. (Eds.). (2018). *Human resource information systems* (4th ed.). Sage.
- Moore, G. (2011). *Systems of engagement and the future of enterprise IT*. AIIM.
- Tornatzky, L. G., & Fleischer, M. (1990). *The processes of technological innovation*. Lexington Books.